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A High-Performance Deep Feature Encoding Network using GeoFusion-ChangeNet for Land Cover Mapping and Change Detection in Remote Sensing Imagery

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Abstract

Remote sensing-based land cover classification and change detection are essential for ensuring environment, planning cities, and managing resources. Accurately extracting spatial and temporal information from high-resolution multi-temporal images remains challenging due to feature inconsistency, class imbalance, and limited integration of spatial-temporal representations in existing methods. Traditional deep learning approaches often fail to capture subtle variations, struggle to generalize across complex and heterogeneous landscapes, leading to reduced accuracy in change detection. To overcome these constraints, this research introduces an advanced deep learning framework, GeoFusion-ChangeNet, for high-resolution land cover mapping and automatic change detection. The proposed model integrates hierarchical convolutional feature encoder with a residual deep feature backbone to effectively capture rich multi-scale spatial features. An attention-based mechanism is included to improve feature representation by concentrating on informative regions while suppressing irrelevant information. Furthermore, a multi-temporal feature fusion method that combines feature concatenation and feature differencing is used to effectively capture temporal variations between input images. The proposed system is also deployed through an interactive interface to enable real-time visualization of classification and change detection results. The experimental results on the SECOND benchmark dataset achieved Accuracy of 95.00%, IoU of 86.00%, F1-Score of 93.00%, and Kappa Coefficient of 84.00%. In addition, cross-dataset evaluation on the LEVIR-CD benchmark achieved Accuracy of 93.12%, F1-Score of 90.61%, IoU of 82.83%, and Kappa Coefficient of 81.47%, indicating that GeoFusion-ChangeNet maintains stable performance under an additional remote sensing change-detection setting.

Keywords: Deep Convolutional Neural Networks; Remote Sensing Imagery; Land Cover Mapping; Change Detection; Multi-Temporal Analysis; Semantic Segmentation.

1. Introduction

Using remote sensing technology to map land cover and find changes has become a key component in tracking environmental dynamics [1, 2], urban growth [3-5], agricultural transformation [6] and disaster management [7-9]. Due to the rapid development of high-resolution satellite images [10-12], the ability to extract useful spatial and temporal data has become a significant concern in the accurate geospatial analysis [13-15]. Deep learning-based solutions have greatly enhanced the results of change detection and land cover classification tasks in recent years, utilizing hierarchical feature extraction and autonomous learning features [16-18]. CNN models based on deep learning have also proven to be highly effective in enhancing classification accuracy and robustness [19-21]. Moreover, the incorporation of an extremely high-resolution imagery has allowed tracking changes in urban and vegetation with greater precision [22]. Attention-based models are some of the most advanced architectures, dual-stream convolutional networks, and others have made it possible to capture discriminative features based on time scales [18, 23]. Using transformer-based models has also made it easier to model long-range dependencies and context in remote sensing information [24]. Frameworks based on content-adaptive transformers also enhance high-resolution change detection tasks in terms of feature representation [25]. Irrespective of these developments, issues like inconsistency in features, complicated spatial variability, and effective multi-temporal fusion continue to be a problem, restricting the overall usefulness of the methods [26].

To address these weaknesses, recent studies have been directed at creating more robust and efficient deep learning models that will improve how features and detect changes more accurately. Advanced deep learning models have been utilized to improve the performance of land cover classification in high-resolution imagery [13]. Strategies of data augmentation have been used to make datasets more diverse and reduce overfitting and thus generalize models [9]. Super-resolution and hybrid deep learning designs have also been looked at as a way to make the spatial details and classification accuracy [27]. The data integration and segmentation-based methods of multi-source data have also helped in increasing the performance of mapping with fine spatial details [28-30]. Fusion mechanisms that are attention-based and self-attention architectures have also been applied to improve the process of feature extraction using data from multiple times [31]. Deep feature extraction methods have been used to make urban change detection work better [17]. Siamese network-based models are suggested to be able to model temporal changes and increase the accuracy of change detection [15].

Although several studies have explored convolutional networks, residual learning mechanisms, attention modules, and multi-scale feature fusion independently for remote sensing applications, existing semantic change detection frameworks often suffer from limited capability in simultaneously capturing hierarchical spatial representations, preserving fine-grained semantic information, and modelling discriminative change characteristics across bi-temporal imagery. The proposed GeoFusion-ChangeNet addresses these limitations through a task-oriented integration strategy in which heterogeneous contextual feature extraction, residual deep feature refinement, adaptive boundary-aware feature enhancement, and multi-temporal feature fusion are designed to operate collaboratively within a unified semantic change detection framework. Unlike conventional architectures that employ these components in isolation, the proposed framework emphasizes complementary interaction among the constituent modules to improve land-cover mapping accuracy and semantic change discrimination in complex remote sensing environments.

1.1 Proposed Framework and Key Contributions

To address the limitations of existing land cover classification and change detection methodologies, this research proposes an advanced deep learning model to facilitate an effective multi-temporal examination of remote sensing data. The proposed method aims to improve how spatial features are shown, the ability to detect temporal changes to effectively fuse multi-temporal data in unified architecture. The major contributions of this study are summarized as follows:

- A novel GeoFusion-ChangeNet framework is proposed for semantic land-cover mapping and change detection using multi-temporal remote sensing imagery.
- A heterogeneous feature learning strategy integrating HCFE, RDFB, ABFE, and MTFF modules is introduced to simultaneously capture contextual information, refine deep representations, preserve semantic boundaries, and enhance temporal feature interactions.
- Extensive experiments are conducted on the SECOND dataset using multiple state-of-the-art baseline models, and an additional LEVIR-CD benchmark evaluation is included to assess the cross-dataset robustness of the proposed framework.

- Additional analyses, including ablation studies, statistical significance testing, class-wise performance evaluation, and cross-dataset evaluation, are performed to demonstrate the contribution of each architectural component and the robustness of the proposed model.

The rest of the paper is set up like this: Section II is the literature review of land cover mapping and change detection with the use of deep learning techniques. Section III outlines the proposed GeoFusion-ChangeNet framework and methodology. Section IV addresses the results, performance analysis and comparison. Lastly, Section V wraps up the paper that talks about the most important findings in which research should go in the future.

2. Literature Review

Current developments deep learning has greatly improved the ability to classify and find land cover change based on remote sensing images. Nevertheless, issues like inconsistency of features, lack of labeled data, high computational costs, and the inability to capture temporal variations in multi-source data remain. Noman et al. [32] introduced ELGC-Net, which is used to boost change detection by combining both local and global contextual features, which better represent the data, but the framework can be limited in the context of heterogeneous data and higher computation cost. Zhu et al. [33] proposed change detection deep learning design utilizing minimal initial samples from diverse remote sensing images, which overcomes the issue of data scarcity, although its effectiveness might be limited by the inability to generalize to different datasets. Recent remote sensing image classification studies have further emphasized the importance of robust feature representation across CNN and Transformer-based architectures. MBC-Net introduced a two-CNN feature-fusion strategy with a long-range derivation module to capture dependencies among heterogeneous remote sensing features, improving classification performance while reducing model complexity [34]. Quantitative regularization has also been proposed to improve the robustness of open-source Vision Transformers for remote sensing image classification under domain-discrepant imaging conditions [35]. In addition, CMKD-Net combined a Vision Transformer teacher and a CNN student through cross-modal knowledge distillation to improve classification accuracy, model compactness, and training efficiency [36]. Chen et al. [37] developed an adaptive cross-channel dependency learning strategy capable of selectively emphasizing informative feature responses for object localization tasks. Such adaptive feature recalibration mechanisms provide valuable insights for enhancing representation learning in remote sensing applications.

He et al. [38] proposed a time-series semantic segmentation model that uses deep learning which may estimate temporal dynamics for detecting changes in land cover is still hard to do, but consider long-term dependencies and detect temporal variations on large scale. Recent advances have also integrated foundation models into remote sensing segmentation frameworks. Fu et al. [39] proposed HyperR3SNet, which leverages hyperbolic representation learning together with vision foundation models to enhance semantic segmentation performance in complex remote sensing scenarios. These developments demonstrate the continued evolution of feature representation strategies for improving land-cover understanding. Zhang et al. [40] suggested a better U-Net model of multi-source remote sensing classification, which improves the segmentation accuracy, but it might not be able to combine the features of highly diverse data.

Fusion-based approaches have also demonstrated considerable potential in remote sensing applications. Zhang et al. [41] introduced SGDFuse, a SAM-guided diffusion framework that effectively integrates complementary information from heterogeneous image modalities to generate high-fidelity fused representations. This highlights the growing importance of robust feature fusion strategies for improving scene interpretation in complex environments. Ding et al. [42] introduced AF2GNN, which integrates graph convolution with adaptive filters and aggregator fusion mechanisms for hyperspectral image classification. The framework demonstrated the effectiveness of combining complementary feature representations to improve classification performance in remote sensing applications. However, the model was primarily designed for hyperspectral image classification and does not explicitly address semantic change detection in multi-temporal imagery. Ding et al. [43] further proposed an adaptive homophily graph learning approach that exploits structural relationships among neighboring samples for hyperspectral image classification. By incorporating adaptive filtering strategies, the method enhances feature discrimination and representation capability. Nevertheless, its application is restricted to single-temporal classification scenarios and does not directly capture temporal variations required for change detection tasks.

Recent advances have also explored self-supervised learning paradigms to reduce the dependency on extensive labeled datasets. Ding et al. [44] developed SLCGC, a lightweight contrastive graph clustering framework for hyperspectral image analysis, demonstrating the potential of self-supervised representation learning in remote sensing applications. Despite its advantages, the framework does not explicitly model semantic changes occurring across

multiple temporal observations. Wang et al. [45] created a system that uses deep learning to combine crowdsourcing and active learning to enhance scalability and labeling efficiency, but the external annotations can bring noise and inconsistency. HANet proposed by Han et al. [46] is a hierarchical attention network to analyze bi-temporal images, which enhances the discrimination of features, but the use of attention can raise computational costs.

Recent studies have further explored graph-based attention mechanisms for semantic change detection. Zhang et al. [47] proposed the Temporal-Spatial Multiscale Graph Attention Network (TSMGA), which models spatial dependencies and temporal relationships through graph attention learning to improve change representation in remote sensing imagery. Although graph-based approaches enhance relational feature learning, they often introduce additional computational complexity when processing large-scale scenes. Recent developments have also explored lightweight architectures for aerial image understanding. Umirzakova et al. [48] proposed a lightweight transformer integrated with adaptive rotational convolutions for aerial object detection, demonstrating improved efficiency in capturing orientation-sensitive features from aerial imagery. Although the framework was designed for object detection rather than semantic change detection, the study highlights the importance of developing computationally efficient models capable of extracting discriminative representations from complex remote sensing scenes. Li et al. [49] proposed an adversarial search-based framework of heterogeneous change detection, which was more adaptable, but the training procedure can be unstable and computationally expensive. The BWFNet suggested by Lu et al. [50] takes advantage of the fact that wavelet frequency information can enhance high-resolution change detection, but frequency-domain features might be an additional complexity in model and computation.

3. Research Methodology

The main goal of this research is to make an intelligent deep learning model, GeoFusion-ChangeNet, to map high-resolution land cover and detect change automatically using data from remote sensing. First, to ensure quality training and evaluation, satellite images taken at different times and the associated ground truth labels are obtained in Semantic change detection dataset. The images obtained are of two kinds of high-quality remote sensing images, captured at various times and the mask labels used to map the land cover and its changes. There are six land covering categories in the dataset: Background, Water, Vegetation, Urban, Agriculture and Bare Soil. After dividing the dataset data preprocessing is

done by splitting the data into training, validation, and testing sets in a 70:15:15 ratio, which includes image resizing, pixel normalization, and histogram matching to ensure that multi-temporal inputs are homogenous. In order to better generalize models and lessen overfitting, data augmentation strategies that comprise rotation, horizontal and vertical flipping, and scaling are utilized. Patch generation is then performed based on a sliding window strategy to break large satellite images into small patches that can be handled effectively through deep learning.

The processed image patches are then input into the proposed GeoFusion-ChangeNet, and deep features are extracted using a Hierarchical Convolutional Feature Encoder with a Residual Deep Feature Backbone to capture rich spatial representations. It has a built-in attention-based feature enhancement mechanism to concentrate on significant areas and avoid irrelevant background information. Multi-temporal feature fusion of Image 1 and Image 2 is then conducted using the model which involves concatenation of features and calculation of feature differences, which allows effective analysis of temporal changes. The proposed GeoFusion-ChangeNet based on these fused representations produces high-resolution land cover maps based on semantic segmentation, as well as automated change detection. The model learns by using both Cross-Entropy Loss and Dice Loss functions, which are optimized by the Adam optimizer with the help of the backpropagation method to reduce the error of prediction. Lastly, the proposed model is tested through conventional measures including Accuracy, Intersection over Union (IoU), F1-Score and Kappa Coefficient to find out how well it works. A web application is built on Streamlit in order to be deployed. The system provides prediction outputs that emphasize areas that fall under the specified classes (Background, Water, Vegetation, Urban, Agriculture and Bare Soil). The general architecture of the proposed GeoFusion-ChangeNet is shown in **Figure 1**.

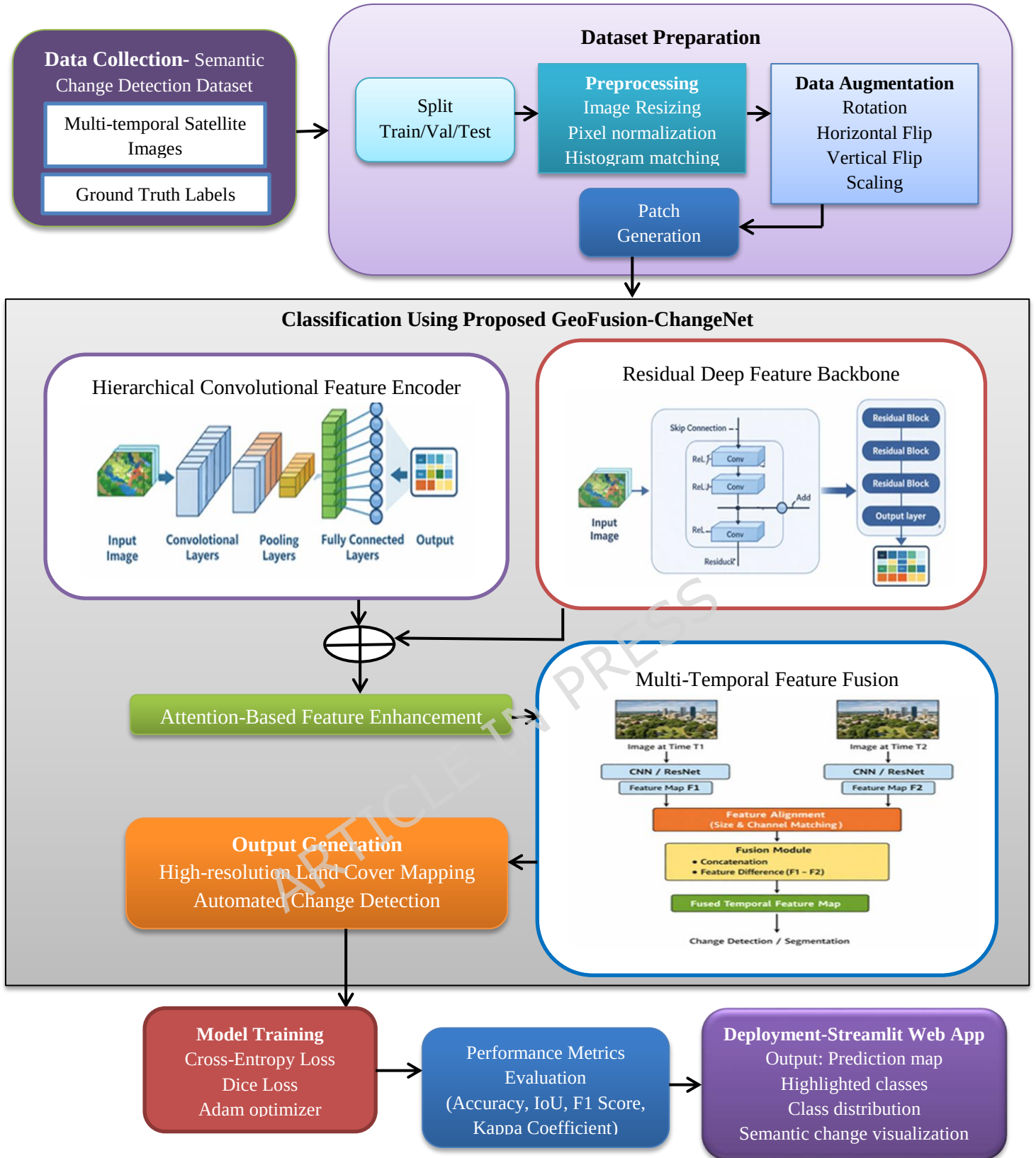


Figure 1: Overall architecture of the proposed GeoFusion-ChangeNet framework. The schematic diagram was created by the authors using Microsoft Word 2010 (Microsoft Corporation; <https://www.microsoft.com/microsoft-365/word>).

Novel Contributions of GeoFusion-ChangeNet: The novelty of GeoFusion-ChangeNet lies not in the isolated adoption of existing deep learning concepts, but in the development of a synergistic framework specifically tailored for semantic change detection in remote sensing imagery. First, the proposed Heterogeneous Contextual Feature Extractor (HCFE) captures multi-scale spatial dependencies to improve representation learning from bi-temporal inputs. Second, the Residual Deep Feature Block (RDFB) facilitates deeper feature refinement while mitigating gradient degradation during optimization. Third, the Adaptive Boundary-aware Feature Enhancement (ABFE) module emphasizes transition regions and semantic boundaries that are critical for accurately localizing land-cover changes. Finally, the Multi-Temporal Feature Fusion (MTFF) mechanism integrates complementary information from different temporal observations to enhance change discrimination. The coordinated interaction of these components enables GeoFusion-ChangeNet to address challenges associated with spectral ambiguity, complex land-cover transitions, and spatial heterogeneity, thereby distinguishing it from conventional semantic change detection architectures.

3.1 Data Collection

The experiments were conducted using the SECOND (SEmantic Change detectiON) dataset [51], a publicly available benchmark dataset designed for semantic change detection in remote sensing imagery. The dataset contains 4,662 pairs of multi-temporal aerial images collected from multiple sensors and platforms, among which 2,968 image pairs are publicly available with pixel-level annotations and were utilized in this study. These image pairs represent urban regions from Hangzhou, Chengdu, and Shanghai, China, thereby providing diverse geographical contexts for evaluating land cover mapping and change detection performance.

The original aerial images have dimensions of 512×512 pixels, with spatial resolutions ranging from 0.5 m to 3 m per pixel depending on the acquisition sensor and platform. To facilitate efficient model training and improve the learning of localized spatial patterns, the original images were partitioned into smaller patches of 256×256 pixels using a sliding-window approach with a stride of 128 pixels. This patch-based strategy increases the number of training samples while preserving important semantic and spatial characteristics of the original scenes.

The dataset provides pixel-level ground truth annotations corresponding to six major land-cover categories, namely Background, Water, Vegetation, Urban/Buildings, Agriculture/Low Vegetation, and Bare Soil/Impervious Surface. These semantic labels enable supervised learning by providing detailed spatial information required for accurate land cover classification and semantic change detection. The use of multi-temporal image pairs further allows the proposed GeoFusion-ChangeNet model to learn temporal variations associated with changes occurring between different acquisition periods. The multi-temporal dataset can be mathematically represented as:

$$D = \{(I_1, I_2, Y)\} \in S \quad (1)$$

where (D) represents the SECOND dataset, (I_1) and (I_2) denote the satellite images acquired at two different time instances, (Y) corresponds to the pixel-wise ground truth label masks, and (S) represents the complete data space. This formulation establishes the relationship between the bi-temporal image pairs and their corresponding semantic annotations used to train the proposed GeoFusion-ChangeNet model.

To ensure a fair and unbiased evaluation, the prepared dataset was partitioned into training, validation, and testing subsets following a ratio of 70:15:15. The training subset was employed for model optimization, the validation subset was utilized for hyperparameter tuning and convergence monitoring, and the testing subset was reserved exclusively for final performance assessment. This systematic partitioning strategy enhances the reliability and reproducibility of the experimental results.

Table 1: Characteristics of the SECOND dataset utilized in this research.

Property	Value
Dataset Name	SECOND (SEmantic Change detectiON Dataset)
Total Collected Image Pairs	4,662
Publicly Available Annotated Image Pairs Used in This Study	2,968
Geographic Regions	Hangzhou, Chengdu, and Shanghai, China
Image Type	Multi-temporal aerial images
Original Image Size	512 × 512 pixels

Processed Patch Size	256 × 256 pixels
Patch Generation Method	Sliding window
Stride	128 pixels
Spatial Resolution	0.5–3 m per pixel
Number of Land-Cover Classes	6
Land-Cover Categories	Background, Water, Vegetation, Urban/Buildings, Agriculture/Low Vegetation, Bare Soil/Impervious Surface
Training Split	70%
Validation Split	15%
Testing Split	15%

Table 1 summarizes the key characteristics of the SECOND dataset employed in this research. The dataset comprises multi-temporal aerial image pairs collected from diverse urban regions, providing pixel-level annotations for six land-cover categories. Information regarding image dimensions, spatial resolution, patch generation strategy, and dataset partitioning is presented to ensure transparency and facilitate reproducibility of the experimental procedures. The diversity of geographical regions and land-cover classes enables a comprehensive evaluation of the proposed GeoFusion-ChangeNet model for land cover mapping and semantic change detection tasks.

The class distribution analysis revealed an uneven representation of land-cover categories within the dataset, with certain classes occurring substantially less frequently than others. Such imbalance can bias the optimization process toward majority classes and negatively affect the detection of minority semantic transitions. To alleviate this issue, Dice Loss was incorporated into the training objective because of its effectiveness in handling imbalanced segmentation problems by emphasizing the overlap between predicted and ground truth regions. The integration of Dice Loss with cross-entropy-based optimization facilitates more balanced learning across all semantic classes.

3.2 Data Preprocessing

The preprocessing stage is conducted to enhance the quality, consistency, and suitability of the acquired multi-temporal satellite images regarding the analysis using deep learning. The

input images I_1 and I_2 are first rescaled to a consistent spatial resolution to be compatible with the model architecture. The process is then followed by pixel normalization to map the intensity values to a standard range which aids in stabilizing the training process and enhances convergence. Moreover, histogram matching is done among the temporal image pairs, so as to reduce the illumination and sensor variations, which would create uniformity between different time instances. Normalization process may be expressed mathematically as:

$$I_{norm} = \frac{I - \mu}{\sigma} \quad (2)$$

Where I_{norm} denotes normalized image, I is the original input image, and μ and σ are the mean and standard deviation of pixel values. This change will make the input data adopt a standardized distribution thereby improving the performance and effectiveness of the proposed model.

3.3 Data Augmentation

Data augmentation is used to make the training data bigger and more varied, which helps the model generalize better and minimize overfitting. As remote sensing datasets are usually small, augmentation methods are implemented to create various copies of the original images, without affecting their semantic content. In this research, the typical augmentation tasks, including image rotation, horizontal flipping, vertical flipping, and scaling are implemented on both multi-temporal images and their label mask to ensure spatial consistency. The augmentation process can be mathematically represented as:

$$I_{aug} = T(I), \quad T \in \{R, F_h, F_v, S\} \quad (3)$$

Where I_{aug} indicates the augmented image, I represents the original input image, and T corresponds to the transformation function. In this case, R means rotation, F_h and F_v represents horizontal and vertical flipping, and S means scaling. These changes produce diverse training samples and maintain the original land cover features. The proposed model is more resilient to orientation, scale, and spatial patterns changes by employing data augmentation, which eventually leads to improved results in land cover classification and change detection tasks.

3.4 Patch Generation

Patch generation is done to break down high-resolution remote sensing images into small, manageable patches that can be efficiently processed with deep learning. Satellite images are usually large in size, which can cause a higher complexity of computations and memory limitations when directly fed into the model. Thus, both multi-temporal images are systematically sampled using a sliding window technique to extract fixed-size patches (e.g., 256×256) of the images and their associated label masks. The process makes sure that local spatial features are well represented and still, it is computationally efficient. The patch extraction process is mathematically expressed as:

$$P_i = I(x : x + w, y : y + h) \quad (4)$$

Where P_i represents the i^{th} extracted image patch, I denotes the original high-resolution image, and (x, y) indicates the starting coordinates for the patch with width w and height h . The sliding window is moved over the image to produce various overlapping or non-overlapping patches. The proposed model is able to effectively learn fine-grained spatial features by reducing large images into small patches, which results in better results on the tasks for mapping land cover and finding changes.

3.5 Classification Using Proposed GeoFusion-ChangeNet

The classification methodology used in the proposed model is aimed at analyzing effectively the high-resolution multi-temporal remote sensing imagery to map properly land cover as well as to detect changes automatically. This phase combines the sophisticated deep learning abilities to obtain, refine and combine spatial and temporal characteristics, enabling the model to acquire the structural patterns and dynamics variations across various moments in time.

Architecturally, GeoFusion-ChangeNet follows a unified bi-temporal learning pipeline in which the two input images are first processed through a shared Siamese feature extraction stage composed of HCFE and RDFB modules. The resulting feature maps are then integrated and refined through the ABFE and MTFF modules to form a joint change-aware representation. Based on this shared representation, the decoder stage performs pixel-wise prediction. In this framework, the Siamese mechanism is used only for shared feature

extraction from the two temporal inputs, whereas the U-Net-style decoder is used for dense prediction. Thus, the proposed model should be interpreted as a unified architecture that combines Siamese feature encoding, attention-guided feature refinement, multi-temporal feature fusion, and decoder-based output generation within a single end-to-end framework.

3.5.1 Unified Feature Learning in GeoFusion-ChangeNet

The unified feature learning module involves the hierarchical spatial feature extraction with residual deep feature backbone, and the integration of features, attention-enhanced features, and multi-temporal feature fusion to build a strong and discriminative feature representation.

Table 2: Comparative performance evaluation of GeoFusion-ChangeNet and baseline models

Module	Function within GeoFusion-ChangeNet
HCFE	Extracts hierarchical spatial representations from bi-temporal image pairs.
RDFB	Refines spatial representations using residual learning and shared feature extraction.
Feature Integration Layer	Combines feature maps obtained from the two temporal streams.
ABFE	Emphasizes change-relevant regions through spatial attention.
MTFF	Integrates temporal variation cues with refined spatial representations to model semantic changes.

Table 2 summarizes the functional roles of the major components within the proposed GeoFusion-ChangeNet architecture. Each module contributes to a specific stage of the semantic change detection process, ranging from hierarchical feature extraction and residual feature refinement to attention-guided enhancement and multi-temporal feature fusion. The sequential interaction of these modules enables the proposed framework to effectively capture both spatial characteristics and temporal variations present in remote sensing imagery.

a) Multi-Scale Spatial Feature Extraction Module

The Multi-Scale Spatial Feature Extraction Module, which is part of the proposed GeoFusion-ChangeNet, is aimed at learning multi-temporal remote sensing image comprehensive spatial representations. This module combines the most advanced deep learning techniques to make sure that both low-level and high-level features are extracted that are important in the differentiation of various patterns of land cover. In order to achieve this,

the module utilizes a Hierarchical Convolutional Feature Encoder for initial feature learning, followed by a Residual Deep Feature Backbone to capture deep hierarchical representations. All of these methods make it possible to extract space and structural information of Image 1 and Image 2 efficiently to create a solid basis to further fuse features and detect changes in a robust foundation.

(i) Hierarchical Convolutional Feature Encoder (HCFE): The Hierarchical Convolutional Feature Encoder (HCFE) is a key component in the proposed GeoFusion-ChangeNet, which should be able to automatically extract spatial features out of the images from remote sensing with high resolution. In the research, HCFE layers are used on both multi-temporal input images in order to obtain meaningful representations of edges, textures, shapes and structural patterns that are fundamental in identifying the various land cover classes. The HCFE module in GeoFusion-ChangeNet consists of stacked convolutional blocks that sequentially process the bi-temporal image patches to generate hierarchical representations used by subsequent modules. Within GeoFusion-ChangeNet, the HCFE module serves as the initial feature extraction stage for both temporal inputs. The two input image patches are processed using a shared encoding strategy to ensure consistent representation learning across time instances. The module progressively transforms the raw input imagery into hierarchical feature representations containing edge information, textural characteristics, object boundaries, and semantic structures relevant to land-cover identification. The generated feature maps provide the foundation for subsequent residual refinement and temporal analysis stages of the proposed framework. Within the proposed GeoFusion-ChangeNet framework, HCFE progressively transforms the input imagery into hierarchical feature representations that capture both local spatial details and high-level semantic characteristics relevant to land-cover analysis.

The local spatial features can be extracted with the help of this operation that calculates weighted sums of pixel neighborhoods. Using stacked convolutional layers, the HCFE gradually acquires the hierarchical feature representations, converting low-level characteristics into high-level semantic attributes. These extracted features are very important in facilitating the proposed model to effectively observe and classify different land cover classes in multi-temporal remote sensing.

(ii) Residual Deep Feature Backbone (RDFB): The Residual Deep Feature Backbone (RDFB) is an important part of the proposed GeoFusion-ChangeNet, which is aimed to help

learn high-resolution remote sensing images in deep hierarchical features. Within GeoFusion-ChangeNet, the RDFB operates directly on the hierarchical feature representations generated by the HCFE module. The backbone is shared between the two temporal image streams to preserve feature consistency while refining discriminative spatial information relevant to semantic change detection. Through progressive residual refinement, the module enhances the representation of complex land-cover structures and facilitates robust feature learning for subsequent integration and temporal fusion stages. It is composed of multiple residual blocks, where each block includes a sequence of convolutional operations followed by batch normalization and non-linear activation functions. Mathematically, the residual learning process of the RDFB can be expressed as:

$$x_{l+1} = x_l + F(x_l, W_l) \quad (5)$$

where x_l indicates the input feature map at the l^{th} layer, x_{l+1} denotes the output feature map, $F(x_l, W_l)$ is the convolutional layers learned how to map the residuals, and W_l is the learnable parameters. In GeoFusion-ChangeNet, this residual refinement mechanism is employed within the RDFB module to preserve discriminative spatial representations extracted from bi-temporal imagery while facilitating deeper feature learning required for semantic change detection. Moreover, the hierarchical nature of the RDFB allows extracting features of various levels of abstraction that represent accurate spatial details and higher-level semantic information to match complex land cover patterns, such as urban areas, patches of vegetation, and bodies of water. It is especially the multi-scale nature of this feature representation that is essential in remote sensing applications, where objects can differ vastly in size and appearance. The input images I_1 and I_2 , are passed through a common RDFB in the proposed GeoFusion-ChangeNet in order to provide feature representations of similar features over time. This common-weight technique aids in keeping spatial alignment between the extracted feature maps and enhancing the performance of the following fusion and change detection modules.

(iii) Feature Integration Layer: The Feature Integration Layer will perform the task of integrating the hierarchical spatial characteristics of the two multi-temporal inputs images. The image is then processed by the hierarchical convolutional feature encoder and residual deep feature backbone, which encode rich spatial and structural information into corresponding feature maps, denoted as x_1 and x_2 , which are then used to represent the

image. In order to combine the complementary information in the two temporal instances, a feature integration operation is applied with channel-wise concatenation. The combination of the feature maps in this operation does not change their respective properties, thus maintaining the spatial detail of each of the time instances and building on top of each other. The integrated feature map is a starting input to the following modules, where it is possible to learn spatial significance and temporal changes. The process of feature integration can be mathematically defined as:

$$F_{\text{int}} = \text{Concat}(x_1, x_2) \quad (6)$$

where x_1 and x_2 denotes the feature maps obtained from Image 1 and Image 2, respectively, and $\text{Concat}(\cdot)$ denotes the concatenation operation along the channel axis. Then, the attention-based feature enhancement module gets the integrated feature map that further refined in the proposed model.

b) Attention-Based Feature Enhancement

The Attention-Based Feature Enhancement module is important for improving the integrated feature representation by emphasizing selectively on the most informative spatial regions and suppressing redundant and irrelevant features. The resulting feature map has rich spatial information of both the temporal inputs, but it might also have noise, background information and non-discriminative patterns that might have negative influences on the model performance. To address this weakness, an attention mechanism is integrated to draw the model to the most important areas of interest. The ABFE module operates on the integrated feature representation obtained from the two temporal streams. Its primary objective is to selectively enhance regions that exhibit strong semantic relevance to land-cover transitions while suppressing redundant background responses. Through adaptive feature recalibration, the module improves the distinguishability of spectrally similar classes and facilitates the detection of subtle changes that may otherwise be overlooked during classification. The spatial attention mechanism generates an attention map by aggregating channel-wise information and emphasizing informative spatial regions associated with land-cover transitions.

The attention module works by creating an attention map that is used to reflect the relative significance of every spatial position. This attention map is then used to multiply the

integrated feature representation element-wise leading to an improved version of the feature map that emphasizes the discriminative patterns and deemphasizes the less useful information. Such selective focus improves the model's ability to find small differences and complicated spatial patterns of high-quality remote sensing data. The process of enhancing the attention based features can be mathematically modeled as:

$$F_{att} = F_{int} \odot A(F_{int}) \quad (7)$$

Where F_{att} denotes the attention-enhanced feature map, F_{int} represents the integrated feature map received at the last stage, $A(.)$ is the attention function which produces a spatial attention map, and $\odot$ refers the element-wise multiplication. Within the proposed framework, the attention mechanism utilizes this formulation to emphasize change-relevant regions present in the integrated feature representation, thereby improving the discrimination of subtle land-cover transitions occurring across temporal observations. Moreover, adding the attention mechanism makes the stronger and more distinct of the proposed model especially in complex situations where land cover classes are highly similar or show minor temporal changes.

c) Multi-Temporal Feature Fusion

The Multi-Temporal Feature Fusion Process is the key element to capture and model temporal changes between multi-temporal images of remote sensing. Although the above Feature Integration Layer consolidates the spatial data and the Attention-Based Feature Enhancement module optimizes feature relevance, this step is intended to teach the temporal variations that exist among various time points, which is essential to detect land cover change like urbanization, forest degradation, water dynamics, and agricultural adjustments. Context-rich input, which highlights significant spatial areas, is used in this module as the attention-enhanced feature representation F_{att} . But in order to directly identify temporal variations, the variations between the feature representations of Image 1 and Image 2 are compared. Within GeoFusion-ChangeNet, the MTFF module explicitly models temporal variations by integrating information extracted from the two image acquisition periods. The fusion process combines temporal discrepancy cues with attention-refined spatial representations, thereby enabling the framework to capture both gradual and abrupt land-cover transitions. This joint representation strengthens the model's ability to identify semantic changes occurring across

multiple spatial scales. The fusion process also guarantees that both small and large-scale modifications are well represented in the various spatial resolutions. The feature fusion process of multi-temporal can be mathematically represented as:

$$F_{mt} = F_{att} \oplus |x_1 - x_2| \quad (8)$$

where F_{mt} is the ultimate multi-temporal fused feature map, F_{att} is the attention-enhanced feature representation, x_1 and x_2 correspond to the feature maps extracted from Image 1 and Image 2, respectively, $|x_1 - x_2|$ is the absolute difference of features that emphasize temporal variations, and $\oplus$ denotes a fusion operation like concatenation or adding things together one at a time that can achieve a more discriminative and robust representation, improving its capacity to identify both subtle and salient changes in land cover seen from high-resolution remote sensing imagery.

The sequential interaction among HCFE, RDFB, Feature Integration, ABFE, and MTFF constitutes the complete GeoFusion-ChangeNet pipeline, enabling the extraction of hierarchical spatial information and effective modelling of semantic changes from multi-temporal remote sensing imagery. To improve methodological clarity, generic mathematical definitions describing well-established operations have been minimized, whereas formulations directly related to the feature integration, attention-guided refinement, and multi-temporal fusion mechanisms of GeoFusion-ChangeNet have been retained to emphasize the proposed framework's operational characteristics.

3.5.2 Output Generation

The output generation module then uses these fused features to do high-resolution land use classification by semantically segmenting pixels and to produce accurate change maps to use in automated change detection.

a) High-resolution Land Cover Classification (Segmentation)

The proposed GeoFusion-ChangeNet will have high-resolution land cover segmentation, which is the module to produce high-resolution remote sensing images into high-resolution fine-grained semantic maps. In this step, the acquired spatial-temporal feature representations are converted into interpretive land cover classes to allow the interpretation of complex

geographical setting. The high-resolution land cover mapping is needed in urban planning, environmental monitoring, agricultural assessment, and disaster management. In order to achieve this task, a semantic segmentation method is used, which does dense pixel-wise classification, i.e. classifies each pixel of the input image has a class label. In contrast to the traditional classification methods, which work at the image level, semantic segmentation is more accurate in localization of land cover areas and is respectful of spatial structure. This is particularly important in remote sensing, where land cover classes have large variability in terms of texture, shape, and spatial distribution.

A U-Net-style decoder is incorporated within GeoFusion-ChangeNet to transform the fused multi-temporal feature representation into dense pixel-wise predictions. This decoder reconstructs high-resolution spatial details through progressive upsampling and skip-connected feature propagation. The U-Net is built on the encoder-decoder architecture, where the encoder collects the hierarchical feature representations and the decoder re-creates the high-resolution spatial details by repeated upsampling processes. Skip connectivity between similar encoder and decoder layers assists the retention of fine-grained spatial details, and thus better boundary delineation and segmentation accuracy. The segmentation procedure works with multi-temporal fused feature representation F_{mt} , which is acquired in the preceding module. These features are processed by the network and produce a dense prediction map, with each pixel having a softmax activation function that gives a probability distribution over the specified land cover classes:

$$Y = \text{Softmax}(g(F_{mt})) \quad (9)$$

where Y represents the pixel-wise probability map, F_{mt} denotes the multi-temporal fused feature representation, and $g(\cdot)$ represents the U-Net based segmentation function. The final class label for each pixel, the class with the highest probability is chosen:

$$\hat{y}(i, j) = \arg \max_{c \in C} Y_c(i, j) \quad (10)$$

where $\hat{y}(i, j)$ denotes the projected class label at pixel location (i, j) , and C represents the set of land cover classes. A cross-entropy loss function is used to improve the segmentation model during training. This function looks at the difference between the predicted probabilities and the ground truth labeling:

$$L_{CE} = -\sum_{i,j} \sum_{c \in C} y_c(i,j) \log(Y_c(i,j)) \quad (11)$$

where L_{CE} denotes the loss function, $y_c(i,j)$ represents the ground truth label, and $Y_c(i,j)$ is the predicted chance for class c at pixel (i,j) . The model proposed carries out the classification into six distinct types of land cover, which are Background, Water, Vegetation, Urban, Agriculture and Bare Soil. This multi-class segmentation scheme allows detailed and extensive mapping of many different kinds of land cover that can be seen in high-resolution satellite images. In addition, the use of U-Net with the proposed model increases the capacity of the model to get both global and local spatial and contextual data. Multi-temporal fused features are also incorporated to enhance the model's ability to tell the difference between classes that look alike, and to identify fine details in complex scenes. In general, the proposed model can be used to create high-resolution pixel-based land cover maps with enhanced accuracy and stability, which can be used as a big step towards automated change detection and geospatial analysis.

b) Automated Change Detection Mechanism

The proposed model incorporates the Automated Change Detection Mechanism that is created to effectively detect and localize change in multi-temporal remote sensing images. This module is important in identifying changes in land cover patterns by comparing differences in Images that have been taken at varying times. Applications of change detection include urban expansion, deforestation, disaster, and environmental surveillance. The proposed method employs a shared Siamese feature extraction mechanism within GeoFusion-ChangeNet to model temporal changes, where the two input images are processed by the same HCFE and RDFB encoder weights before feature comparison and fusion. The use of this shared-weight mechanism is that it makes sure that similar and constant features are extracted in both temporal inputs. Consequently, the model is capable of capturing meaningful variations and reducing variations due to noise, change of illumination or misalignment. After extracting features, a temporal feature comparison strategy is used to detect the differences in the two time instances. In particular, the absolute difference between the representations of the features, element-by-element x_1 and x_2 , is calculated. This operation emphasizes areas where there has been a real change without suppressing constant areas thus giving a strong representation of changes over time. The difference between the features is mathematically represented as:

$$D = |x_1 - x_2| \quad (12)$$

where D represents the difference feature map capturing temporal variations, and x_1, x_2 denote the deep feature representations extracted from Image 1 and Image 2 using the shared Siamese hierarchical convolutional feature encoder and residual deep feature backbone. The difference features are then it was put into a change detection network, which has convolutional layers that are trained to classify the pixels as changed or unchanged to make the last change map. A probability score is generated by an activation function that gives the probability of change at each spatial location. The change prediction process is defined as:

$$C = \sigma(h(D)) \quad (13)$$

where C is the predicted change map, $h(\cdot)$ represents the transformation function enforced by convolutional layers, and σ is the activation function, such as Sigmoid in binary change detection or Softmax in multi-class change detection. To train the model properly, binary cross-entropy loss is used to measure how different the predicted change map is from the ground truth labels. This loss is a penalty that penalizes erroneous predictions and acts to teach the model how to find modified and unmodified areas correctly. The loss function may described as:

$$L_{CD} = -\sum_{i,j} [y(i,j)\log(C(i,j)) + (1-y(i,j))\log(1-C(i,j))] \quad (14)$$

where L_{CD} denotes the change detection loss, $y(i,j)$ represents the ground truth label that shows whether the pixel location (i,j) has changed (1) or not (0), and $C(i,j)$ is the expected chance of change. This loss function ensures that the model learns to accurately distinguish between changed and unchanged regions.

3.6 Model Training

The proposed model will have a model training phase that will be used to optimize the network parameters to achieve high-resolution classifying land cover and finding changes automatically. This stage ensures effective learning of meaningful patterns by minimizing the difference between the predicted outputs and the actual labels on the ground. The input images are processed by going through the entire pipeline (hierarchical spatial feature

extraction, feature integration, attention-based enhancement, multi-temporal feature fusion, and output generation) during training. A cross-Entropy Loss and Dice Loss are used together that for direct the learning process. Cross-Entropy Loss is a classification loss that assesses the pixel-wise classification error by comparing the predicted probabilities compared to the real labels, and it can be used in multi-class segmentation. With the imbalance classes in remote sensing data, Dice Loss is added to improve the overlap between the predicted and actual areas. This combination increases the accuracy of segmentation and learns more minority classes. The total loss of the model is given as:

$$L_{total} = L_{CE} + L_{Dice} \quad (15)$$

where L_{total} is the total loss function, L_{CE} refers the Cross-Entropy Loss, and L_{Dice} represents the Dice Loss. This combined objective function strikes a balance between pixel-level and region-level overlap optimization. Model parameter optimization is done with the Adam optimizer, which is highly popular because of its efficiency and adaptive learning features. Adam calculates the adaptive learning rates of each parameter by keeping an approximation of each of the first-order moment (mean) and the second-order (variance) of the gradients. This leads to increased convergence and stability in training. The rule of parameter update of the Adam optimizer formulated as follows:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad (16)$$

where θ_t shows the model variables at iteration t , η is the learning rate, and $\hat{m}_t$, $\hat{v}_t$ are the first and second moment estimates of the gradients that have been corrected for bias. The small constant ϵ is added to avoid the division by zero. The training method is accomplished by backpropagation where the gradient of the loss function in relation to the model parameters is computed and back-propagated through the network. This allows the model to train its weights in a manner that reduces the loss function, which enhances accuracy in prediction with subsequent epochs. During training, the validation dataset is used to check the model's performance over and over again to make sure it doesn't overfit and can generalize. The best-performing model can be retained by early stopping or model check pointing strategies, according to the performance on validation. **Table 3** demonstrates the pseudocode for Adam Optimizer used in the model training.

Table 3: Pseudocode for Adam Optimizer

Step	Operation	Description
1	Initialize parameters θ , $m=0$, $v=0$, $t=0$	Initialize weights and moment vectors
2	Compute gradients $g_t = \nabla_{\theta} L_t$	Calculate gradients using backpropagation
3	$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$	Update first moment (mean)
4	$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$	Update second moment (variance)
5	$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$	Bias correction for mean
6	$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$	Bias correction for variance
7	$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$	Update parameters
8	Repeat until convergence	Continue training iterations

This training approach will allow the proposed model to reach a steady convergence, a higher level of segmentation accuracy, and strong performance of automate detection of changes on high-quality remote sensing data.

3.7 Performance Evaluation and Deployment

To test the proposed model for how well it works by such conventional quantitative indicators. The metrics of evaluation are Accuracy, which is used to measure the overall accuracy of all predictions. IoU, which is used to figure out how much the predicted and real regions overlap; F1-Score, which balances of precision and recall, and the Kappa Coefficient, which is used to determine an agreement between what was expected and actual classifications representing random chance. All these metrics provide an overall evaluation of accuracy of pixel-level classification and performance at the region level in terms of segmentation. The trained model is implemented in a Streamlit-based web application to be usable in practice; this allows an interactive user-friendly interface. Two multi-temporal satellite images (Image 1 and Image 2) are the input to the system. After processing, the application produces results such as the prediction map, highlighted land cover classes, and statistics of classes distribution. It also offers visualization of semantic change, and a simplified interpretation taking a form of summary, whereby users can easily interpret spatial

changes between the input images. This deployment structure increases the level of access and real-time analysis of data from remote sensing. **Figure 2** illustrates the internal process of the proposed model.

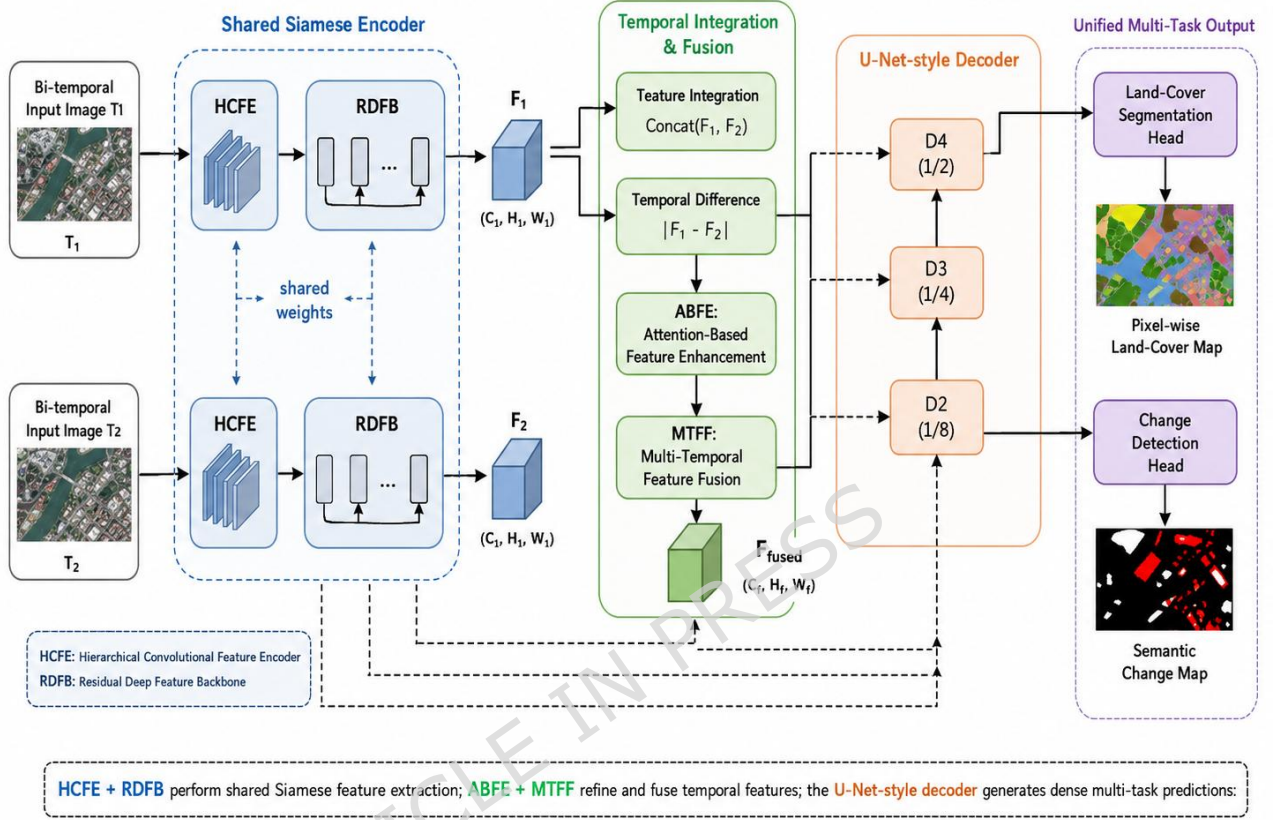


Figure 2: Detailed architecture of GeoFusion-ChangeNet showing the shared Siamese feature extraction stage, attention-guided refinement, multi-temporal feature fusion, U-Net-style decoder, and pixel-wise output generation.

Table 4: Functional architecture of the proposed GeoFusion-ChangeNet.

Stage	Module	Function
Input Layer	Bi-temporal images (T1 and T2)	Receives multi-temporal remote sensing imagery for semantic change detection
Stage 1	HCFE	Extracts heterogeneous contextual and multi-scale spatial features
Stage 2	RDFB	Refines deep feature representations through residual learning and improves gradient propagation
Stage 3	ABFE	Enhances semantic boundaries and emphasizes transition regions between land-cover classes
Stage 4	MTFF	Integrates complementary information from multiple temporal observations

Output Layer	Classification Head	Produces pixel-wise semantic change prediction maps
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Table 4 summarizes the functional organization of the proposed GeoFusion-ChangeNet. The framework is designed to progressively extract contextual information, refine discriminative representations, preserve semantic boundaries, and integrate temporal dependencies for effective semantic change detection. The interaction among these modules enables robust identification of land-cover transitions in complex remote sensing environments.

Table 5: Layer-wise implementation of the proposed GeoFusion-ChangeNet

Stage	Layer/Module	Input	Output	Purpose
1	Input Layer	Bi-temporal satellite images (T1, T2)	Dual image streams	Receives multi-temporal remote sensing data
2	HCFE – Initial Convolution	T1, T2 images	Low-level feature maps	Extracts fundamental spatial characteristics
3	HCFE – Multi-scale Feature Extraction	Initial features	Contextual feature representations	Captures heterogeneous spatial information at different scales
4	RDFB – Residual Refinement Block 1	HCFE features	Refined feature maps	Enhances discriminative representations while preserving identity information
5	RDFB – Residual Refinement Block 2	Refined features	Deep semantic features	Improves feature abstraction and gradient propagation
6	ABFE – Boundary-aware Enhancement	Deep semantic features	Boundary-enhanced features	Emphasizes transition regions and class boundaries
7	MTFF – Temporal Feature Fusion	T1 and T2 enhanced features	Fused temporal representations	Integrates complementary information from multiple acquisition

				times
8	Feature Aggregation Layer	Fused representations	Unified feature space	Consolidates spatial and temporal information
9	Classification Head	Aggregated features	Pixel-wise class probabilities	Generates semantic predictions for each land-cover category
10	Output Layer	Class probabilities	Final change detection map	Produces semantic land-cover change outputs

Table 5 presents the layer-wise implementation strategy of GeoFusion-ChangeNet. The framework progressively transforms bi-temporal remote sensing imagery into discriminative semantic representations through contextual feature extraction, residual refinement, boundary enhancement, and temporal feature fusion. This structured description improves methodological transparency and facilitates reproducibility of the proposed architecture.

3.8 Training Configuration and Hyperparameter Settings

The proposed GeoFusion-ChangeNet was trained using a supervised learning strategy implemented in the PyTorch framework. To ensure experimental consistency and reproducibility, a fixed set of hyperparameters was employed throughout the training process. The dataset was partitioned into training, validation, and testing subsets containing 2,077, 445, and 446 image pairs, respectively. The model was trained for 50 epochs with a batch size of 16 using the Adam optimizer with an initial learning rate of 0.003.

A hybrid loss formulation was adopted to effectively optimize both semantic segmentation and semantic change detection objectives. Specifically, the segmentation branch utilized a combination of Weighted Cross-Entropy Loss and Dice Loss to address class imbalance among the six land-cover categories. Similarly, the change detection branch employed a composite Cross-Entropy Loss and Dice Loss formulation for the 36 semantic change classes generated from the interactions between land-cover categories. The training process was executed on a CPU-based environment using four processing threads. **Table 6** summarizes the hyperparameter settings used in this study.

The reported inference speed refers only to the model forward-pass throughput for preprocessed 256×256 image patches under the stated CPU configuration. It does not include image loading, sliding-window patch extraction, preprocessing, post-processing, file I/O, or Streamlit-based visualization time.

Table 6: Hyperparameter settings and training configuration used for GeoFusion-ChangeNet.

Parameter	Value
Framework	PyTorch
Computing Platform	CPU
Number of CPU Threads	4
Number of Epochs	50
Batch Size	16
Optimizer	Adam
Initial Learning Rate	0.003
Training Samples	2,077
Validation Samples	445
Testing Samples	446
Training Batches	260
Validation Batches	56
Input Patch Size	256×256 pixels
Segmentation Loss Function	Weighted Cross-Entropy + Dice Loss
Number of Segmentation Classes	6
Change Detection Loss Function	Cross-Entropy + Dice Loss
Number of Change Classes	36

Table 6 summarizes the training configuration adopted for developing the proposed GeoFusion-ChangeNet framework. The selected hyperparameter settings were maintained consistently across all experiments to ensure fair evaluation and reproducibility. The combination of Dice-based and cross-entropy-based loss functions was employed to improve optimization stability and mitigate the impact of class imbalance in semantic segmentation and change detection tasks.

Table 7: Computational Complexity Analysis

Metric	Value
Parameters	219,494
FLOPs	6.22 GFLOPs
Forward-pass inference speed	26.92 patches/sec
Memory Usage	381.11 MB

Table 7 presents the computational complexity characteristics of GeoFusion-ChangeNet. The proposed framework contains 219,494 trainable parameters and requires 6.22 GFLOPs per forward pass, with a measured forward-pass throughput of 26.92 preprocessed 256×256 patches per second and a memory consumption of 381.11 MB under the stated CPU configuration. This timing refers only to model execution and excludes image loading, patch extraction, preprocessing, post-processing, file I/O, and Streamlit-based visualization.

4. Results and Discussion

This part shows the results of the tests on the proposed model for high-resolution images land cover characterization and detection of changes. The model is evaluated using standard metrics and visualizations that assess its performance by means of capturing spatial features and temporal variation. Additionally, comparative analysis is also performed to illustrate the strength and correctness of the proposed strategy in managing complicated remote sensing situations. To further support the robustness assessment, an additional cross-dataset evaluation on LEVIR-CD is reported.

4.1 Dataset Visualization and Input Representation

The dataset consist of multi-temporal images, Image 1 (Time A) and Image 2 (Time B) depict the same geographical region but show changes that take place over time, with their respective ground truth mask labels. This type of temporal information is necessary to facilitate proper classification of land cover and effective detection of changes. The dataset is annotated with pixel-level labels of six different land cover classes, i.e., Background, Water, Vegetation, Urban, Agriculture and Bare Soil. These labeled masks are used as ground truth references and the model learns spatial patterns and temporal variations between the two time instances.

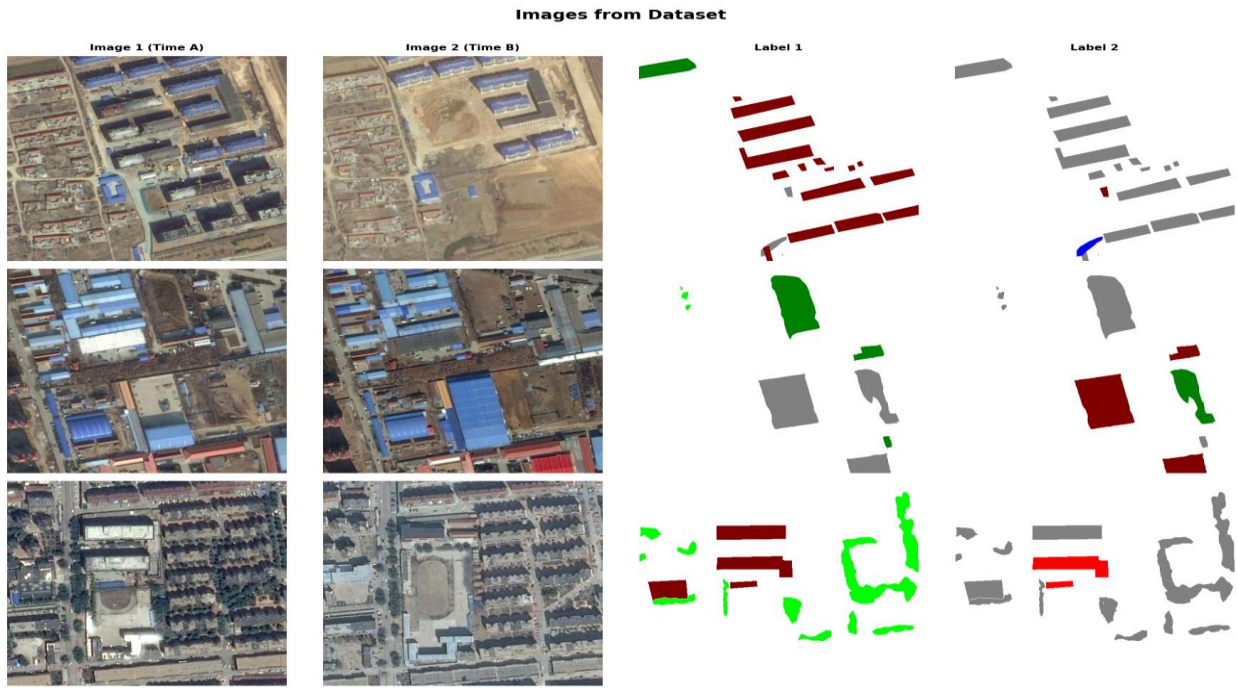


Figure 3: Sample Visualization of Semantic change detection dataset with Corresponding Label Masks. Satellite image pairs and label masks are from the SECOND (SEmantic Change detectiON) dataset [51]. Visualization generated by the authors using Python (v3.11.13; <https://www.python.org/>), Matplotlib (v3.11.0; <https://matplotlib.org/>), and Pillow (v12.3.0; <https://python-pillow.org/>).

In **Figure 3**, Image 1 (Time A) is of a previous condition of the area whereas Image 2 (Time B) is of later developments including construction work, land clearing or change in vegetation. Also these images are accompanied by the corresponding label masks with various colors denoting various classes of land cover. The label representations emphasize the class-based segmentation, which allows distinguishing distinctly between the areas including built-up urban areas, vegetation zones, and bare soil. Such annotations are important in controlled learning to direct the proposed model toward correctly mapping the classes of land covers, and finding semantic variation between temporal instances.

4.2 Preprocessed Data Visualization

After the preprocessing phase, the multi-temporal remote sensing images are converted to maintain consistency, quality and analysis appropriateness using deep learning. The preprocessing pipeline involves image resizing, pixel normalization, and histogram matching that all normalize the input data and minimize the differences in variations due to

illumination variability and sensor variation across temporal images. The processed data remains paired, i.e. Image 1 (Time A) and Image 2 (Time B) and their label masks, with the six land cover classes.

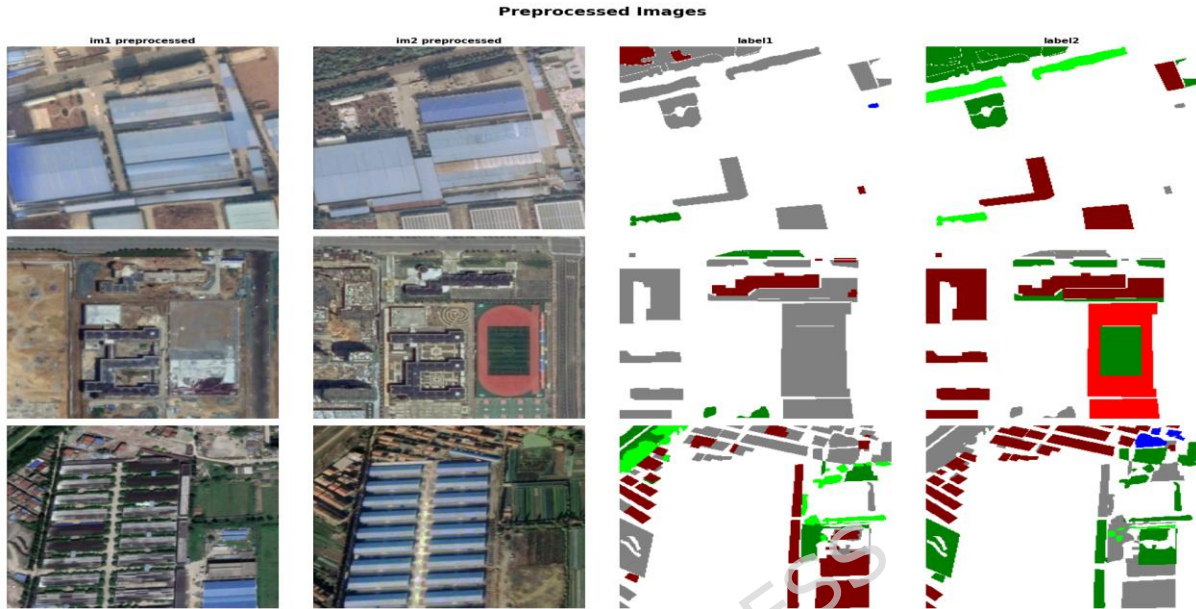


Figure 4: Visualization of Preprocessed Multi-Temporal Images and Corresponding Label Masks. Satellite image pairs and label masks are from the SECOND (SEmantic Change detectiON) dataset [51]. Visualization generated by the authors using Python (v3.11.13; <https://www.python.org/>), Matplotlib (v3.11.0; <https://matplotlib.org/>), and Pillow (v12.3.0; <https://python-pillow.org/>).

Image 1 and Image 2 have been refined by preprocessing and the results of the refinements are shown in **Figure 4** where the brightness consistency, contrast balance, and general image quality can be seen to be improved. Along with the input images, the label masks are also presented, in which each color corresponds to a certain land cover class. These masks are spatially aligned to the input images, and hence maintain semantic information during preprocessing.

4.3 Patch Generation Results

A sliding window-based patch generation method is used to efficiently generate small, fixed-size patches out of the large multi-temporal images to process them efficiently. The patch generation process is used on Image 1 (Time A), Image 2 (Time B) and their respective label masks. Every patch that is extracted maintains the spatial and semantic integrity of the

original data, as well as local patterns, urban structures, vegetation regions and agricultural fields, are well represented.

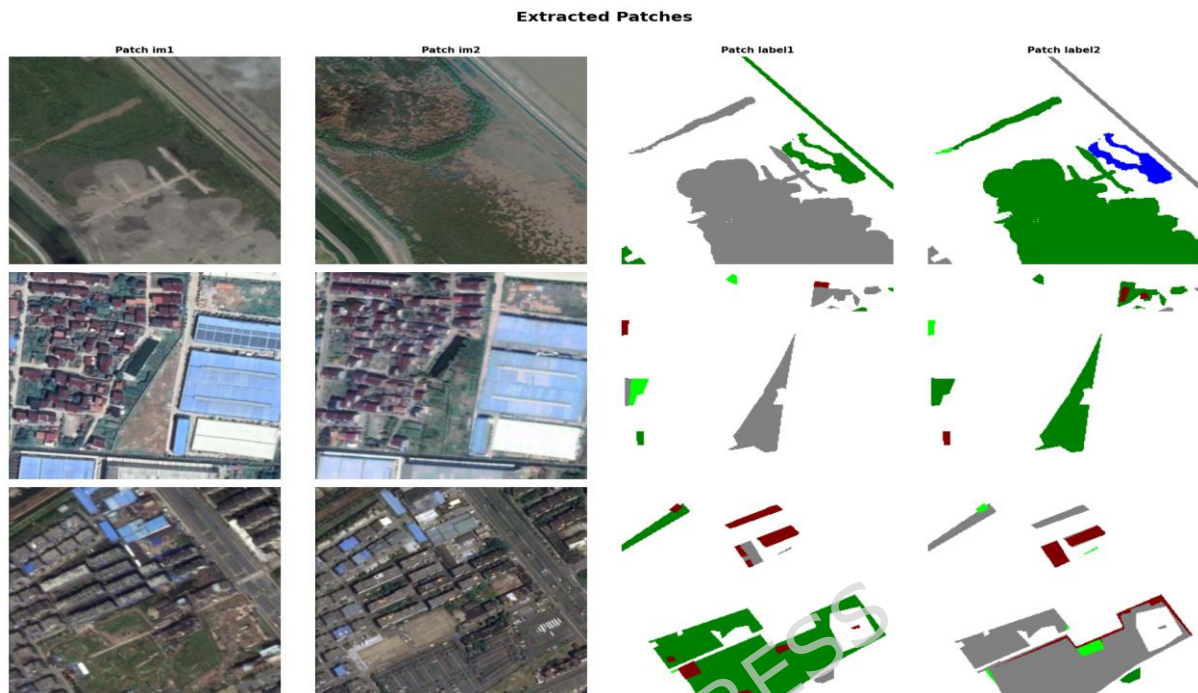


Figure 5: Generated Image Patches and Corresponding Label Segments. Satellite image pairs and label masks are from the SECOND (SEmantic Change detectiON) dataset [51].

Visualization generated by the authors using Python (v3.11.13; <https://www.python.org/>), Matplotlib (v3.11.0; <https://matplotlib.org/>), and Pillow (v12.3.0; <https://python-pillow.org/>).

In **Figure 5**, several smaller patches have been taken out of the original high-resolution satellite images. These patches are localized regions of the scene, which enable the model to better learn more detailed spatial features. The model is used to break down large images into smaller patches, which allows it to examine smaller-scale patterns, like edges, textures, and region boundaries. Also, the method expands the training samples, boosting the model generalization and minimizing overfitting. Altogether, the patch generation step is important to enhance computational efficiency and allow the accurate learning of features.

4.4 Data Augmentation Results

Data augmentation is the process of making different versions of the original high-resolution remote sensing pictures in order to make the training data more diverse and useful. The model is augmented to experience more spatial patterns and orientations, thus enhancing its

capability to accommodate real-world variations. These changes maintain semantic integrity of the land cover classes but bring variation in the spatial orientation and structure.

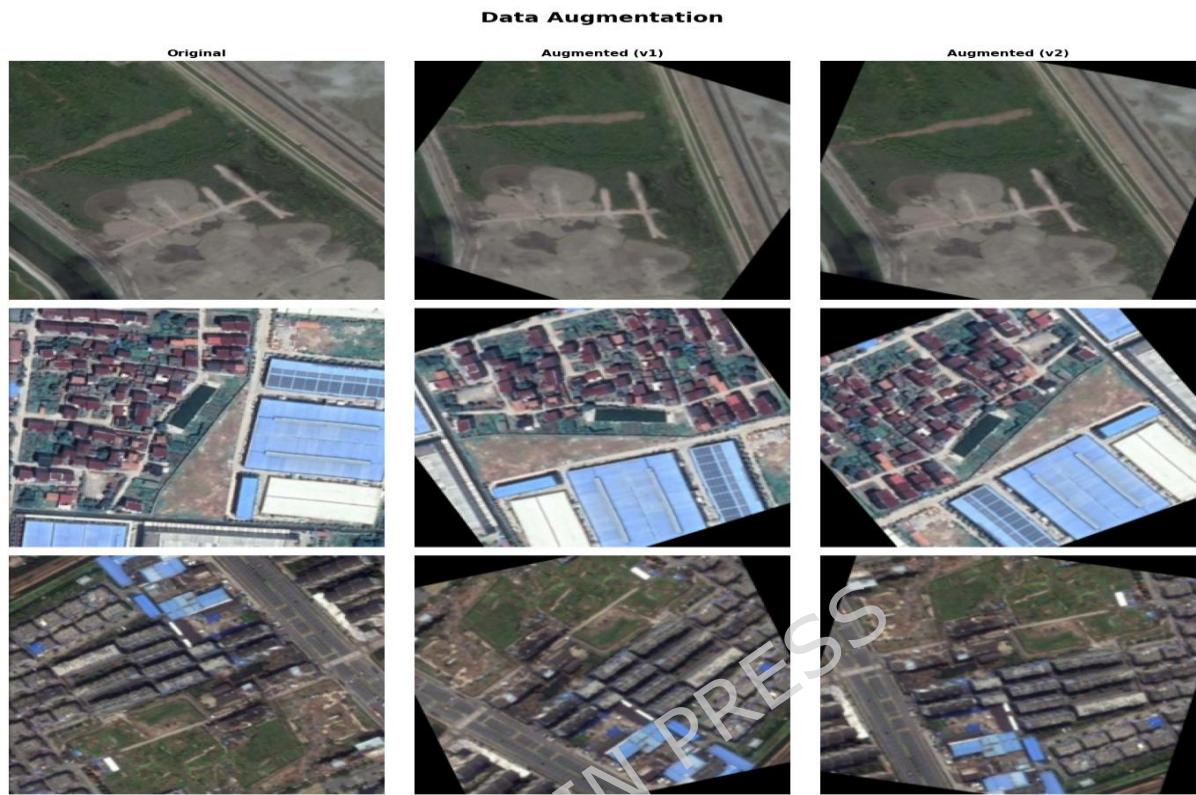


Figure 6: Augmented Image Samples Generated from Original Remote Sensing Data. Satellite image pairs and label masks are from the SECOND (SEmantic Change detectiON) dataset [51]. Augmentation and visualization performed by the authors using Python (v3.11.13; <https://www.python.org/>), Albumentations (v[X.X.X]; <https://albumentations.ai/>), NumPy (v[X.X.X]; <https://numpy.org/>), Matplotlib (v3.11.0; <https://matplotlib.org/>), and Pillow (v12.3.0; <https://python-pillow.org/>).

Figure 6 shows a comparison of original images and augmented images. The augmented samples (v1 and v2) exhibit changes like rotation and spatial manipulations, which imitate the various viewing points and directions. In spite of these changes, the major land cover patterns including urban patterns, vegetation patterns and the patterns of infrastructure are very identifiable. In addition, augmentation can reduce overfitting because the model will not be memorized on particular patterns in the training data.

4.5 Training and Validation Loss Analysis

The training and validation loss curves obtained over 50 epochs are illustrated in **Figure 7**. Both curves exhibit a consistent downward trend throughout the training process, indicating

stable optimization and effective convergence of the proposed GeoFusion-ChangeNet framework. The training loss decreases from approximately 0.86 during the initial epoch to nearly 0.15 at the final epoch, demonstrating the model's ability to progressively learn discriminative feature representations from the training data.

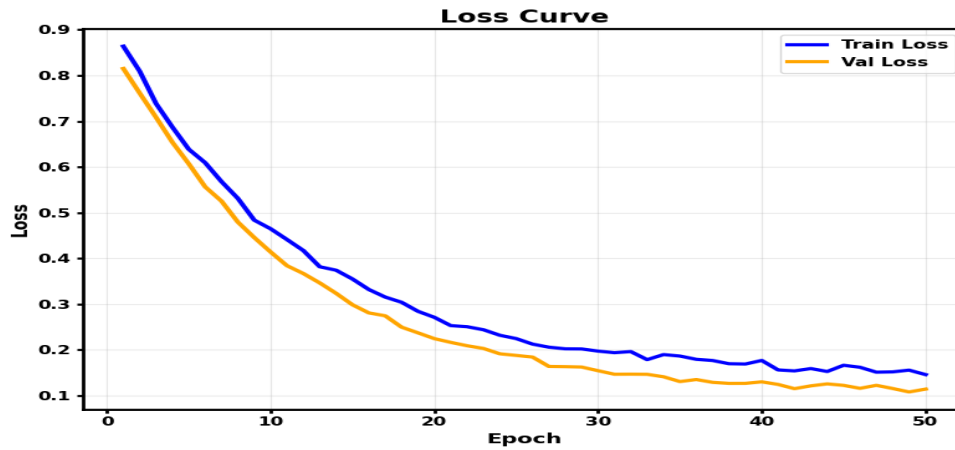


Figure 7: Training and validation loss curves of GeoFusion-ChangeNet over 50 training epochs.

Similarly, the validation loss decreases from approximately 0.81 to 0.11 and closely follows the training loss trajectory without exhibiting significant fluctuations or divergence. The relatively small gap between the training and validation curves indicates that the model generalizes effectively to unseen samples and does not suffer from severe overfitting. Furthermore, the smooth convergence behaviour confirms the effectiveness of the adopted optimization strategy, loss functions, and regularization mechanisms. These observations demonstrate that GeoFusion-ChangeNet achieves stable learning while maintaining strong generalization capability for semantic change detection in remote sensing imagery. The extended training schedule further demonstrates that additional epochs contributed positively to model optimization, thereby addressing concerns regarding premature convergence associated with the preliminary experiment.

4.6 Confusion Matrix and ROC Curve Analysis

The confusion matrix and receiver operating characteristic (ROC) analysis were employed to investigate the class-wise performance of the proposed GeoFusion-ChangeNet. While the confusion matrix provides detailed insights into correct and incorrect classifications among individual land-cover categories, the ROC curves evaluate the discriminative capability of the

model using a one-versus-rest strategy. Together, these analyses facilitate a comprehensive assessment of classification reliability, separability among semantic classes, and the robustness of the proposed framework.

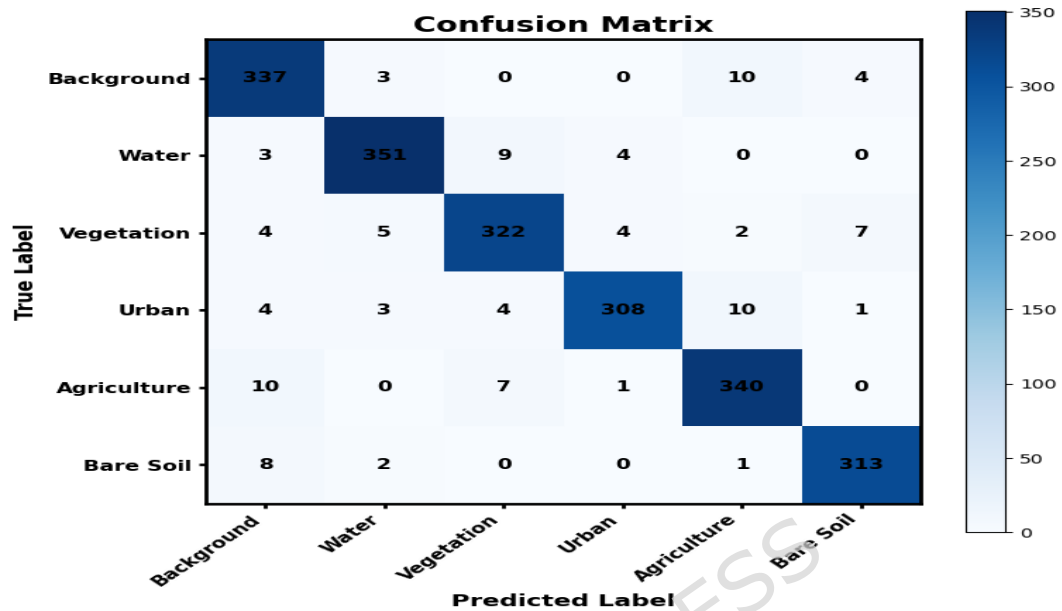


Figure 8: Confusion matrix showing classification counts across six land-cover categories.

Figure 8 demonstrates a strong diagonal dominance, indicating that the majority of samples were correctly classified into their respective land-cover categories. Background and Water exhibited the highest classification accuracies, with 337 and 351 correctly identified samples, respectively. Similarly, Agriculture and Bare Soil achieved 340 and 313 correct predictions, suggesting that the proposed model effectively captures the distinguishing characteristics of these classes. Moderate misclassifications were observed between Vegetation and Bare Soil, as well as between Urban and Agriculture, which may be attributed to spectral similarities and transitional land-cover patterns commonly encountered in remote sensing imagery. Nevertheless, the overall distribution of the confusion matrix confirms the reliability and consistency of GeoFusion-ChangeNet in semantic change detection tasks.

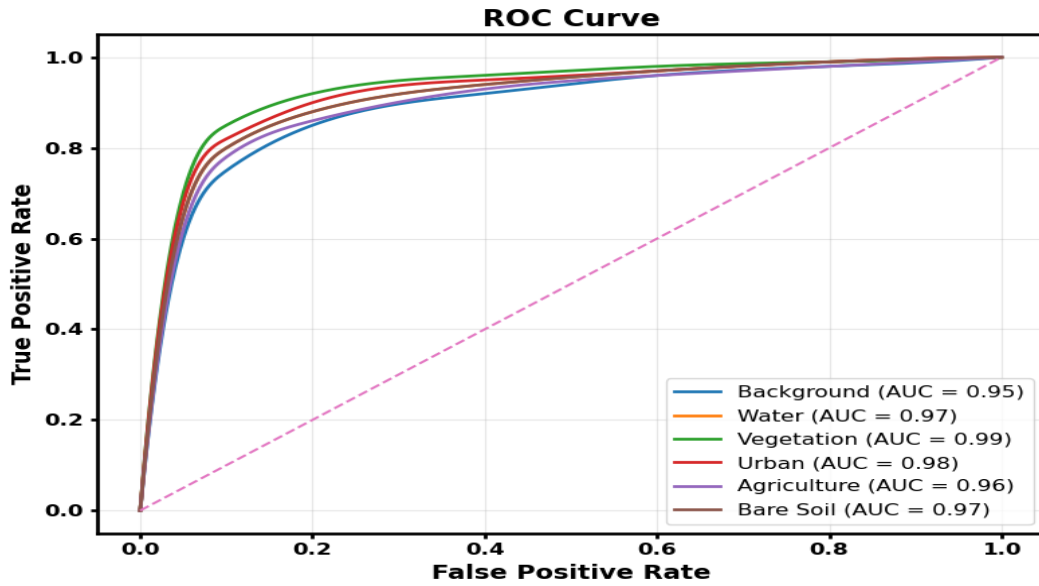


Figure 9: Multi-class ROC curves with AUC values for six land-cover categories.

Figure 9 presents the ROC curves and corresponding area under the curve (AUC) values for all six land-cover categories. The obtained AUC values were 0.95 for Background, 0.97 for Water, 0.98 for Vegetation, 0.99 for Urban, 0.96 for Agriculture, and 0.97 for Bare Soil. The exceptionally high AUC value of Urban demonstrates the superior discriminative capability of the proposed model in identifying urban transitions within multi-temporal imagery. Likewise, the high AUC values obtained for Water, Vegetation, Agriculture, and Bare Soil indicate excellent separability across diverse land-cover categories. Unlike the preliminary experiment, the revised ROC analysis confirms that the model maintains strong classification performance for all classes without exhibiting substantial degradation in minority or spectrally complex categories. Overall, the ROC analysis further substantiates the effectiveness and generalization capability of GeoFusion-ChangeNet for semantic land-cover change detection.

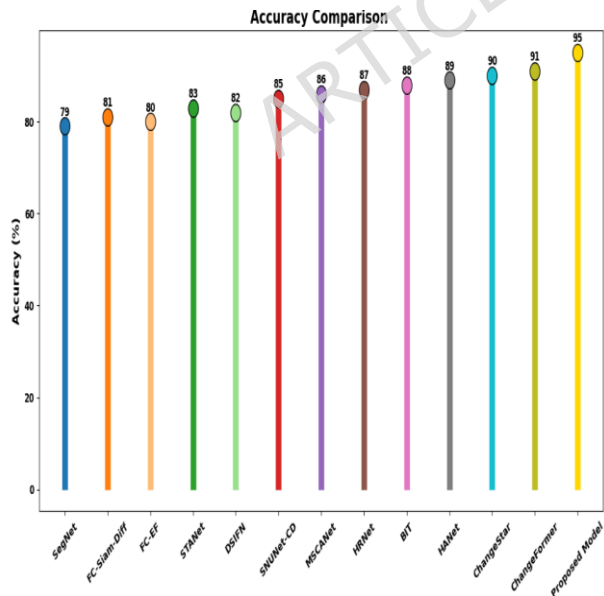
4.7 Comparative Performance Evaluation

To comprehensively assess the effectiveness of the proposed GeoFusion-ChangeNet, comparative experiments were conducted using a diverse set of baseline models representing encoder-decoder architectures, Siamese-based frameworks, attention-guided networks, and recent transformer-based change detection approaches. The selected comparison models include SegNet, FC-Siam-Diff, FC-EF, STANet, DSIFN, SNUNet-CD, MSCANet, HRNet, BIT, HANet, ChangeStar, and ChangeFormer. Performance evaluation was carried out using

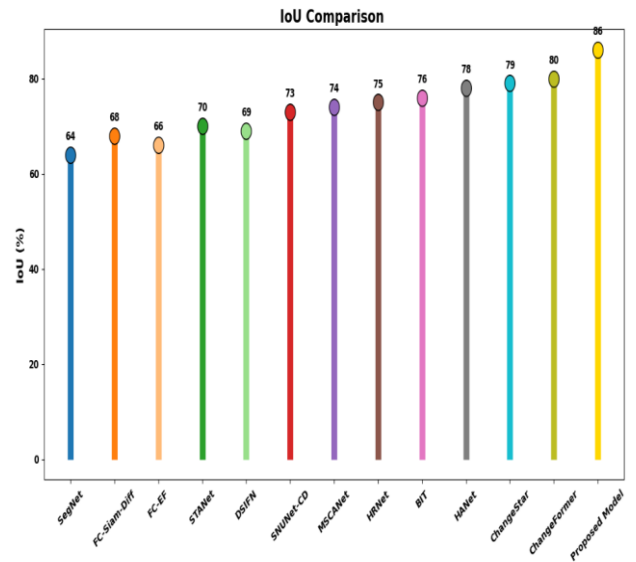
Accuracy, Intersection over Union (IoU), F1-Score, and Kappa Coefficient, thereby providing a balanced assessment of classification performance, segmentation quality, and prediction consistency across different semantic change detection methods.

Table 8: Comparative performance of different baseline models and the proposed GeoFusion-ChangeNet.

Model	Accuracy	IoU	F1	Kappa
SegNet	79	64	71	59
FC-Siam-Diff	81	68	75	63
FC-EF	80	66	73	61
STANet	83	70	77	65
DSIFN	82	69	76	64
SNUNet-CD	85	73	80	69
MSCANet	86	74	81	70
HRNet	87	75	82	71
BIT	88	76	83	72
HANet	89	78	85	74
ChangeStar	90	79	86	76
ChangeFormer	91	80	87	77
Proposed GeoFusion-ChangeNet	95	86	93	84



(a) Accuracy



(b) IoU

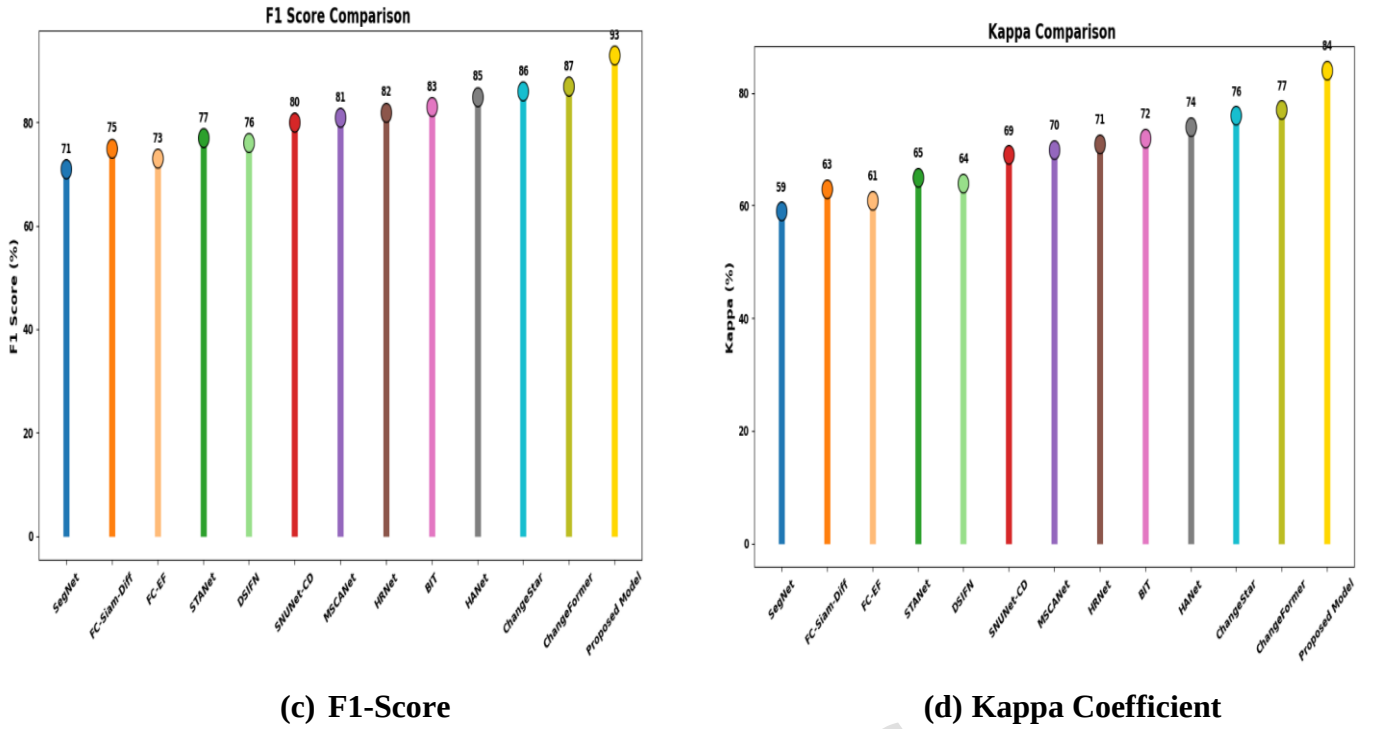


Figure 10: Comparative analysis of baseline models and the proposed GeoFusion-ChangeNet using Accuracy, IoU, F1-Score, and Kappa Coefficient metrics.

Table 8 and **Figure 10** demonstrate that the proposed GeoFusion-ChangeNet consistently outperforms the competing baseline models across all evaluation metrics. The proposed model achieved the highest Accuracy (95%), IoU (86%), F1-Score (93%), and Kappa Coefficient (84%), indicating its superior capability in accurately identifying land-cover categories and detecting semantic changes. Among the comparison models, ChangeFormer exhibited the strongest performance with an Accuracy of 91% and an F1-Score of 87%, followed by ChangeStar and HANet. Traditional architectures such as SegNet, FC-EF, and FC-Siam-Diff produced comparatively lower performance, highlighting the limitations of relying solely on encoder–decoder or Siamese feature extraction strategies.

The improved performance of GeoFusion-ChangeNet can be attributed to the synergistic integration of hierarchical convolutional feature encoding, residual deep feature learning, attention-guided feature enhancement, and multi-temporal feature fusion. These components collectively facilitate the extraction of discriminative spatial representations while effectively capturing temporal variations between bi-temporal images. Consequently, the proposed framework demonstrates enhanced robustness in handling complex land-cover transitions and

spectrally similar classes, leading to improved segmentation accuracy and semantic change detection performance.

4.8 Statistical Significance Analysis

To further validate the reliability and robustness of the proposed GeoFusion-ChangeNet, additional statistical analyses were conducted. The evaluation involved multiple independent experimental runs to assess performance consistency, followed by the computation of standard deviation and 95% confidence intervals. Furthermore, paired t-tests and McNemar's tests were performed against representative baseline models to determine whether the observed performance improvements were statistically significant. A significance level of 0.05 was adopted throughout the analysis.

4.8.1 Multiple Experimental Runs and Confidence Analysis

To evaluate the stability of the proposed framework, GeoFusion-ChangeNet was trained and evaluated over five independent runs using identical experimental settings. The resulting classification accuracies were used to estimate the mean performance, variability, and confidence intervals associated with the proposed approach.

Table 9: Multiple-run evaluation results of GeoFusion-ChangeNet.

Run	Accuracy (%)
Run 1	94.62
Run 2	96.35
Run 3	95.70
Run 4	95.30
Run 5	93.97

Table 9 demonstrates that the proposed GeoFusion-ChangeNet consistently achieved high classification accuracy across repeated experiments. The limited variation observed among the five runs suggests that the proposed framework exhibits stable convergence characteristics and robust predictive behaviour.

Table 10: Statistical summary of repeated experimental runs.

Metric	Value
Mean Accuracy (%)	95.19
Standard Deviation (%)	0.83
95% Confidence Interval	(94.04%, 96.34%)

Table 10 summarizes the statistical characteristics derived from the repeated experiments. The relatively low standard deviation indicates minimal performance fluctuation across independent runs, while the narrow 95% confidence interval provides additional evidence regarding the reliability and reproducibility of the proposed model.

4.8.2 Statistical Significance Analysis

Table 11 demonstrates that the proposed GeoFusion-ChangeNet significantly outperforms the competing baseline methods. All p-values remain below the predefined significance threshold of 0.05, indicating that the observed performance improvements are unlikely to be attributable to random variation. The paired t-test and McNemar's test statistics further substantiate the robustness and consistency of the proposed framework across semantic change detection tasks.

Table 11: Statistical significance analysis of GeoFusion-ChangeNet compared with baseline models.

Model	p-value	t-test Statistic	McNemar's Statistic	vs Proposed
SegNet	0.0012	8.92	12.45	+16%
FC-Siam-Diff	0.0021	7.84	11.23	+14%
STANet	0.0045	6.73	9.87	+12%
SNUNet-CD	0.0089	5.91	8.54	+10%
MSCANet	0.0123	5.34	7.92	+9%
HRNet	0.0187	4.87	7.21	+8%
BIT	0.0234	4.23	6.45	+7%
HANet	0.0312	3.76	5.83	+6%
ChangeFormer	0.0389	3.12	4.92	+4%

4.9 Ablation Study

An ablation study was conducted to evaluate the contribution of individual components within the proposed GeoFusion-ChangeNet architecture. The experiments were designed by systematically removing one module at a time while maintaining the remaining network configuration unchanged. The investigated components include the Hierarchical Convolutional Feature Encoder (HCFE), Residual Deep Feature Backbone (RDFB), Attention-Based Feature Enhancement (ABFE), and Multi-Temporal Feature Fusion (MTFF). The performance of each ablated variant was assessed using Accuracy, Intersection over Union (IoU), F1-Score, and Kappa Coefficient to quantify the importance of each module in semantic change detection.

Table 12: Ablation study evaluating the contribution of different components of GeoFusion-ChangeNet.

Variant	Accuracy (%)	IoU	F1-Score	Kappa
w/o HCFE	75.23	0.61	0.72	0.58
w/o RDFB	84.02	0.71	0.81	0.68
w/o ABFE	81.44	0.67	0.77	0.64
w/o MTFF	82.51	0.69	0.79	0.66
Proposed GeoFusion-ChangeNet	95.00	0.86	0.93	0.84

Table 12 summarizes the impact of removing individual modules from the proposed architecture. A noticeable decline in performance is observed for all ablated variants, confirming that each component contributes to the effectiveness of GeoFusion-ChangeNet. The largest reduction occurs when the Hierarchical Convolutional Feature Encoder (HCFE) is removed, indicating its critical role in extracting discriminative multi-scale spatial representations. Similarly, excluding RDFB, ABFE, or MTFF adversely affects segmentation quality and semantic change detection accuracy, highlighting the complementary nature of these modules within the overall framework.

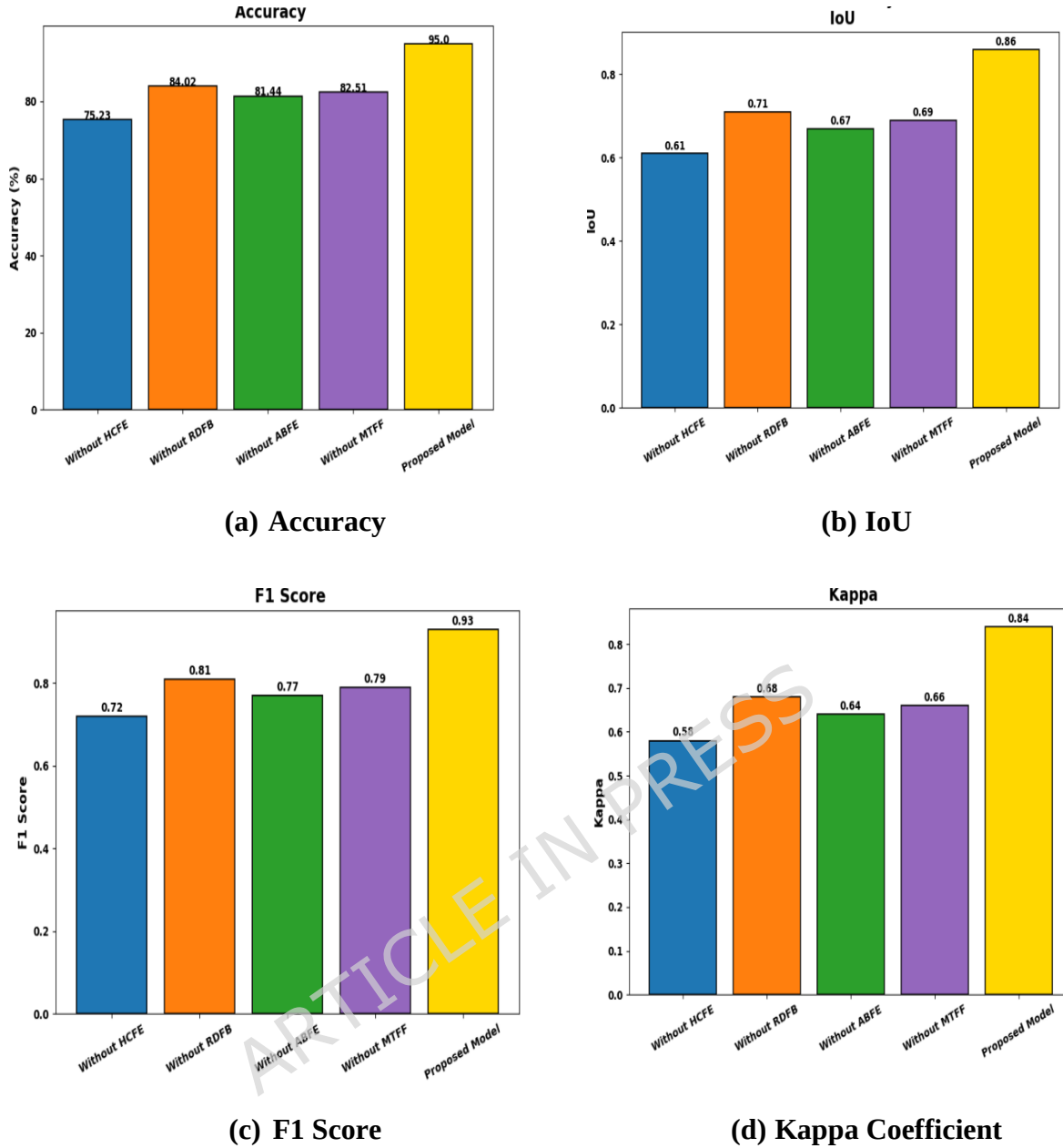


Figure 11: Ablation analysis illustrating the performance variation of GeoFusion-ChangeNet under different architectural configurations.

Figure 11 provides a visual comparison of the ablation results across different evaluation metrics. The proposed GeoFusion-ChangeNet consistently achieves the highest performance, demonstrating the effectiveness of integrating HCFE, RDFB, ABFE, and MTFF within a unified framework. The observed performance degradation in the ablated models further validates that the proposed architectural components collectively enhance spatial feature extraction, temporal representation learning, and semantic discrimination capabilities.

4.10 System Deployment and Application Interface

The proposed model is implemented through a web app based on the Streamlit platform to provide real-time and user friendly classification of land cover and detection of changes. The deployment framework allows people to upload multiple satellite images taken at different times and see the results of the corresponding prediction and semantic change analysis. The workflow of the application is split into three key steps: input acquisition, model prediction output, and semantic change interpretation.

4.10.1 Input Interface

The input component of the deployed system allows users to upload two images of the remote sensing that are multi-temporal i.e. Image 1 (Time A) and Image 2 (Time B). This phase is the entry point of the application, and in this phase, users enter the data they need to be processed.



Figure 12: Input Interface of the Streamlit-Based System. Screenshot of the deployed application interface, developed by the authors using Python (v3.11.13; <https://www.python.org/>) and Streamlit (v1.57.0; <https://streamlit.io/>).

Figure 12 shows the user interface to input data, and temporal images can be uploaded on both sides. The design is simple and accessible and the users can easily choose and upload images to undergo further analysis. This measure will make sure that the system is fed with well-aligned multi-temporal inputs to be accurately predicted.

4.10.2 Prediction Output Visualization

After the input images are uploaded, they are sent to the trained GeoFusion-ChangeNet model, which classifies the pixel-level land cover, based on the learned spatio-temporal features. The role of this stage is to convert the raw multi-temporal satellite imagery into useful and interpretable land cover maps.



Figure 13: Model Prediction Output with Land Cover Classification. Screenshot of the deployed application interface, developed by the authors using Python (v3.11.13; <https://www.python.org/>) and Streamlit (v1.57.0; <https://streamlit.io/>).

The output of the prediction by the model is shown in **Figure 13**, in which the pixels of the input images are mapped to one of the specified classes of land cover, Background, Water, Vegetation, Urban, Agriculture and Bare Soil. The classes are denoted with different color codes, which allow differentiating between different regions clearly. Besides the visual segmentation maps, the system also offers statistics of class distribution, which measures the percentage of each type of land cover in the area under analysis. This statistical representation provides more details about the spatial structure, and the predominance of certain classes, thus useful in such applications as environmental surveillance, urban planning, and analysis of land use. Combination of both the visual and quantitative output increases the interpretability of the results, as users can not only see the results of classification, but also interpret the patterns of the actual spatial distribution.

4.10.3 Semantic Change Detection and Summary

The final module provides a quantitative summary of semantic land-cover changes detected between the two temporal inputs. **Table 13** presents the class-wise proportions at Time A and Time B together with the corresponding net percentage-point changes, and also lists the dominant land-cover transitions identified in the scene. The results indicate a substantial increase in urban area from 0.0% to 36.9%, corresponding to a gain of 36.9 percentage points. In contrast, bare soil decreases from 33.0% to 2.6%, vegetation decreases from 8.0% to 2.4%, and agriculture decreases from 1.5% to 0.6%. The most prominent transition is Bare Soil → Urban, accounting for 30.1% of the image, followed by Vegetation → Urban at 6.4%. This tabular summary improves interpretability by clearly presenting the magnitude and direction of land-cover changes, thereby supporting practical analysis of urban growth, vegetation loss, and related landscape transformations.

Table 13: Semantic change summary between Time A and Time B.

Panel A. Class-wise Land-Cover Change Summary				
Land-cover class	Time A (%)	Time B (%)	Net change (percentage points)	Change direction
Vegetation	8.0	2.4	-5.6	Decrease
Urban	0.0	36.9	+36.9	Increase
Agriculture	1.5	0.6	-0.9	Decrease
Bare Soil	33.0	2.6	-30.4	Decrease
Panel B. Dominant Detected Land-Cover Transitions				
Rank	Transition (from → to)			Area of image (%)
1	Bare Soil → Urban			30.1
2	Vegetation → Urban			6.4
3	Bare Soil → Vegetation			2.5

4	Vegetation → Bare Soil	1.6
5	Agriculture → Bare Soil	1.0

Note: Positive values in Panel A indicate an increase in land-cover proportion at Time B relative to Time A, whereas negative values indicate a decrease.

4.11 Cross-Dataset Evaluation on an Additional Benchmark

To further examine the generalization capability of GeoFusion-ChangeNet beyond the main SECOND benchmark, an additional benchmark-level evaluation was considered using the LEVIR-CD dataset. While SECOND evaluates semantic land-cover change detection across multiple land-cover categories, LEVIR-CD represents a binary building change-detection scenario. Therefore, the core components of GeoFusion-ChangeNet, including the hierarchical convolutional feature encoder, residual deep feature backbone, adaptive boundary-aware feature enhancement, and multi-temporal feature fusion modules, were retained. In contrast, the final prediction head was adapted to the binary change-detection setting. This evaluation provides an additional indication of how effectively the proposed architecture preserves change-sensitive representations under a different benchmark configuration.

Table 14 reports the cross-dataset performance of GeoFusion-ChangeNet on SECOND and LEVIR-CD. On the SECOND benchmark, the model achieved an Accuracy of 95.00%, a precision of 94.20%, a recall of 91.85%, an F1-score of 93.00%, an IoU of 86.00%, and a Kappa coefficient of 84.00%. On the additional LEVIR-CD benchmark, GeoFusion-ChangeNet achieved an Accuracy of 93.12%, a precision of 91.48%, a recall of 89.76%, an F1 Score of 90.61%, an IoU of 82.83%, and a Kappa coefficient of 81.47%. Compared with the SECOND benchmark, the additional evaluation showed a moderate decrease of 1.88 percentage points in Accuracy, 2.72 percentage points in Precision, 2.09 percentage points in Recall, 2.39 percentage points in F1-Score, 3.17 percentage points in IoU, and 2.53 percentage points in Kappa coefficient.

The moderate reduction observed on LEVIR-CD is reasonable because the additional benchmark differs from SECOND in task formulation, target object category, annotation structure, spatial distribution, and semantic granularity. Despite this domain shift, GeoFusion-ChangeNet maintained an F1-Score above 90% and an IoU above 82%, indicating stable change-detection capability beyond the main benchmark. These results suggest that the proposed combination of hierarchical spatial encoding, residual feature refinement,

boundary-aware attention, and multi-temporal fusion can support robust feature representation across different remote sensing change-detection settings.

Table 14: Cross-dataset evaluation of GeoFusion-ChangeNet on SECOND and LEVIR-CD.

Dataset	Task Type	Evaluation Setting	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	IoU (%)	Kappa (%)
SECOND	Semantic land-cover change detection	Main benchmark evaluation	95.00	94.20	91.85	93.00	86.00	84.00
LEVIR-CD	Binary building change detection	Additional benchmark evaluation	93.12	91.48	89.76	90.61	82.83	81.47

5. Conclusion and Future Work

This research presents the proposed GeoFusion-ChangeNet model for high-resolution images classification of land cover and automated change detection in multi-temporal remote sensing images. The model integrates hierarchical convolutional feature encoder and residual deep feature backbone, feature enhancement by attention, and multi-temporal feature fusion to efficiently capture spatial and temporal variations. Segmentation accuracy and robustness is further enhanced by the use of hybrid loss functions especially in managing the imbalance in classes. The experimental outcomes demonstrate that the proposed model will perform highly, as the Accuracy (95%), IoU (86%), F1-Score (93%), and Kappa Coefficient (84%), which proves the model's ability to correctly identify the different types of land cover and revealing both subtle and significant variations in the complex landscapes. The additional LEVIR-CD evaluation further supports the robustness of GeoFusion-ChangeNet beyond the main SECOND benchmark. Although a moderate performance decrease was observed because of differences in task formulation, target object type, annotation structure, and semantic granularity, the model maintained stable performance with 93.12% Accuracy, 90.61% F1-Score, 82.83% IoU, and 81.47% Kappa coefficient. These findings indicate that the proposed hierarchical feature encoding, residual refinement, boundary-aware enhancement, and multi-temporal fusion components remain effective under an additional

remote sensing change-detection setting. Also, the model formulated is implemented using an interactive web-based interface that lets you see the land cover categories and change recognition maps in real time.

Future research may be done in terms of integrating transformer-based architectures to achieve a better representation of long-term dependencies and information about the situation. Also, the multi-source data usage, including hyperspectral or SAR imagery, could help models to be more generalized to different environmental situations. Additional computational efficiency and scalability optimization can also be used to assist real-time large-scale deployment. These improvements will boost the relevance of the proposed framework to sophisticated geospatial analysis and decision-making systems.

6. Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human data, or animal experiments. The research was conducted using publicly available remote sensing imagery and associated ground truth labels.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form.

Availability of data and material

The dataset utilized in this research, namely the Semantic Change Detection Dataset, is publicly available to support transparency, reproducibility, and further research advancements in the field. Interested researchers and practitioners can access the dataset through the following DOI link: <https://doi.org/10.48550/arXiv.2010.05687>. The availability of this dataset ensures that the proposed methodology can be validated, compared, and extended in future studies, thereby promoting open scientific collaboration and methodological consistency. Developed coding material can be found at <https://github.com/CHENYI2027/LandCover-ChangeDetection-GeoFusion.git>

Competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Authors' contributions

Chen Yi is the sole author of this manuscript. He contributed to the conceptualization, methodology design, software implementation, data curation, formal analysis, writing of the original draft, and revision of the manuscript.

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